Period lattices and locally toroidal hypertopes

An application of the common theory of marked kernels

Abstract

We recover the two infinite families of Fernandes, Leemans, Piedade and Ivić Weiss by identifying fixed affine groups and their period lattices. The common kernel and coset theorems then give every parameter at once. The proof includes the exact ordered residues, universal presentations, permutation graphs and full type-preserving automorphism groups. We also verify the paper's two finite residue corrections and correct its subsequent presentation claims. The original constructions and principal results remain attributed to their authors.

The source is the twelve-page author edition of [1]. Its Sections 3 and 4 concern the cubic and square families, respectively; Section 5 concerns two finite presentations. We use AL.1–AL.2, AA.ParabolicPeriods, UA.K, UA.SK.Helly and UA.Geometry from the common paper [3]. Their proofs are acquired once and are not repeated. The standard Euclidean triangle and affine Coxeter realizations, and their toroidal parameter conventions, remain explicitly retained classical inputs. The source's preliminary incidence facts retain the attribution given there to Tits and the earlier literature.

All group products use functional composition. Parameters are introduced only after the fixed affine group and its integral translation lattice have been identified. Exact presentation and finite-set calculations are stated where required; they are not replaced by sampling parameter values.

1 The cubic family and its period lattice

Write $a = \rho_0, b = \rho_1, d = \rho_2, c = \rho_3$. Let E be generated by

$$a(x) = (x_2, x_1, x_3), \qquad b(x) = (x_1, x_3, x_2),$$
$$c(x) = (-x_2, -x_1, x_3), \qquad d(x) = (-x_2, -x_1, 1 - x_3).$$

Products act from right to left. Let Q be the signed permutation group in three coordinates, $\epsilon(q) \in \{0, 1\}$ the parity of its number of sign changes, and put $D = \{v \in \mathbb{Z}^3 : \sum v_i \equiv 0 \pmod 2\}$, $A = \{v \in \mathbb{Z}^3 : \sum v_i = 0\}$.

Lemma 1. *The group E is precisely $\{(q, v) \in Q \ltimes \mathbb{Z}^3 : \sum v_i \equiv \epsilon(q) \pmod 2\}$, with translation kernel D. Its marked presentation is*

$$\langle a, b, d, c \,|a^2, b^2, d^2, c^2, (ab)^3, (ad)^2, (ac)^2,$$
$$(bd)^6, (bc)^3, (dc)^2, (c(bd)^2b)^2\rangle.$$

Adjoining $(a(bd)^2b)^s = 1$ gives exactly $E_s = E/sD$, of order $48s^3$.

Proof. The linear maps generate all 48 signed permutations. Each affine map satisfies the parity condition, which multiplication preserves. Direct multiplication gives $u = a(bd)^2b = (1, (1, -1, 0))$; its Q-conjugates are the twelve roots $\pm e_i \pm e_j$, generating D. Hence the translation kernel is D. Surjectivity to Q proves the description of E. All eleven presentation relators hold.

Identification of the abstract presentation is separate. The complete program `cubic/kernel_certificate.g` forms its epimorphism onto the specified 48-element group and computes the actual kernel presentation by Reidemeister–Schreier rewriting and exact presentation simplification. It has three generators f_1, f_2, f_3 and exactly

$$[f_2, f_3], \qquad f_3 f_1^{-1} f_3^{-1} f_1, \qquad [f_1, f_2]$$

as relators. The full preimage words are emitted and independently evaluated by `cubic/certificate.py`. Their translations are $(0,1,1), (1,0,-1), (-1,0,-1)$, of determinant of absolute value 2. Thus this is the whole free abelian kernel, not its abelianization, and its image is D. AL.2 identifies the presented group with E. The exact GAP rewriting and simplification are declared software trust. Conjugating u^s generates exactly the already normal lattice sD, proving the quotient statement and its order. □

2 All parabolics and uniform intersections

Write $E_J = \langle \rho_i : i \in J \rangle$. A mask m denotes the set of positions of its nonzero binary digits. The only infinite proper parabolic is

$$E_{012} = \{(q, v) : q \in S_3 \times \{\pm I\},\ \sum v_i = \epsilon(q)\}, \tag{1}$$

where ϵ is zero on permutations and one on their negatives. Indeed each of a, b, d satisfies this formula, their linear image has order 12, and the conjugates of u by a, b generate A. This proves equality of every fiber. In particular $A \cap sD = sA$.

All other proper parabolics inject into their linear images. Complete closure of the actual affine transformations gives

m	0	1	2	3	4	5	6	8	9	10	11	12	13	14
$\lvert E_J \rvert$	1	2	2	6	2	4	12	2	4	6	24	4	8	48.

The maximal finite groups fix respectively $p = (-1/2, 1/2, 1/2)$ for E_{123}, $p_1 = (0, 0, 1/2)$ for E_{023}, and 0 for E_{013}. The first has linear image Q, and the last is the even-sign subgroup S_4.

Proposition 1. *For every $s \geq 3$ all 256 marked subgroup intersection identities hold in E_s. Its maximal parabolic orders, in omitted-label order $0, 1, 2, 3$, are $48, 8, 24, 12s^2$.*

Proof. For the fourteen finite supports the certificate checks all 196 exact intersections $E_J \cap E_T = E_{J \cap T}$. For any two elements of these groups with equal linear part, each coordinate of the translation difference has absolute value at most 1. Such a difference belongs to sD only if zero for $s \geq 3$, so these meets survive the quotient.

For every finite support J the certificate also checks $Q_J \cap Q_{012} = Q_{J \cap 012}$, and that each actual affine element of E_J with linear part in Q_{012} satisfies (1). These are exactly the elements of $E_{J \cap 012}$, by injectivity of the finite linear images. Any element of the image of E_J also lying in the image of E_{012} has its linear part in this intersection; it therefore already comes from $E_{J \cap 012}$. Identical supports and the full group give the remaining immediate meets. The orders follow from the table and $[A : sA] = s^2$. □

The toroidal maximal sections are exactly $\{3, 6\}_{(2,0)}$ on (c, b, d) and $\{3, 6\}_{(s,0)}$ on (a, b, d), not just maps with their diagrams. The standard Euclidean reflection group $[3, 6]$ has translation lattice A, generated by the conjugates of $u = a(bd)^2 b$; the imposed lattice is exactly sA. This is the usual ordered parameter $(s, 0)$. For the first section the corresponding word with a replaced by c has order two. It is a quotient of $[3, 6]_{(2,0)}$, whose lattice calculation gives order $12 \cdot 2^2 = 48$. Our subgroup has order 48, so the marked epimorphism is an isomorphism. The Euclidean reflection presentation and its torus parameter convention are explicitly retained classical facts. The other maximal sections are C_2^3 and the tetrahedral string group S_4.

3 A faithful action and the literal CPR diagrams

Let

$$w_0 = (1,1,1),\ w_1 = (1,-1,-1),\ w_2 = (-1,1,-1),\ w_3 = (-1,-1,1).$$

The $4s$ labels (i,k), with $k \in \mathbb{Z}/s\mathbb{Z}$, represent plane families $w_i \cdot x = 2k + 1/2$ modulo period $2s$ in the constant. If $qw_i = \eta w_j$, then $\eta = (-1)^{\epsilon(q)}$ is independent of i. An element (q,v) acts by

$$(i,k) \longmapsto \left(j, \eta k + \frac{\eta - 1 + 2w_j \cdot v}{4}\right). \tag{2}$$

The parity condition makes the fraction integral; translation by sD is trivial. The generators a, b, c transpose family pairs $(1,2), (2,3), (0,3)$ respectively, preserving k. The map d swaps families $1, 2$ with $k \mapsto -k-1$, and preserves $0, 3$ with $k \mapsto -k$.

Lemma 2. *For $s \geq 3$ this is a faithful action of E_s, with exactly the two parity-dependent CPR diagrams on [1, p. 6].*

Proof. An element fixing all four family labels has $q = I$ or $-I$: these are the kernel of the signed-permutation action on the four diagonal lines. For $q = -I$, fixing both levels $0, 1$ forces $s \mid 2$, impossible. For $q = I$ triviality says $w_i \cdot v \in 2s\mathbb{Z}$ for every i. The vectors $(w_0 + w_i)/2$, $i = 1, 2, 3$, are the coordinate vectors, so $v \in s\mathbb{Z}^3$. The condition on w_0 then gives $v/s \in D$. Thus the kernel in E is exactly sD.

The following explicit labeling was read against the source graphics. Begin with $(0,0)$–$(3,0)$, an edge of label 3. Append $s-1$ alternating four-vertex blocks, each joined to the preceding last vertex by label 1:

$$\begin{aligned} \text{diamond } h: &\quad (2,h), (1,h), (1,-h-1), (2,-h-1), \\ \text{square } h: &\quad (3,-h-1), (0,-h-1), (0,h+1), (3,h+1). \end{aligned}$$

In each diamond label 0 joins first–second and third–fourth, and label 2 joins first–third and second–fourth. In each square label 3 joins first–second and third–fourth; label 2 joins first–fourth and second–third. For $s = 2t$, finish with label 1 to $(3,t)$ then label 3 to $(0,t)$. For $s = 2t+1$, finish with label 1 to $(2,t)$ then parallel labels $0, 2$ to $(1,t)$. All levels are modulo s. The alternating levels $0, -1, 1, -2, 2, \ldots$ show every label occurs once. The explicit generator formulas give precisely the source's two drawings. This is uniform for both parities; the separately run examples are checks of the transcription, not a proof of the family. IP makes these faithful permutation graphs CPR graphs. □

The range matters: at $s = 2$ the nontrivial element $(-I, (1,1,-1))$ fixes all eight labels. No faithful degree-eight claim follows.

4 Flag transitivity

Let $P_i = (E_s)_{I \setminus \{i\}}$. For three subgroups X, Y, Z, Tits' elementary coset criterion says that

$$(X \cap Y)(X \cap Z) = X \cap YZ \tag{3}$$

is equivalent to the three corresponding coset types having the Helly property. First, each P_i has this local property on its three maximal parabolics. The certificate checks (3) on the finite pair subgroups. For supports $7, 11, 13, 14$ the product sizes are $12, 12, 8, 36$ and the intersection size is 4 in each case. The equal-linear-part translation-difference bounds are $1, 0, 0, 0$, so all four identities survive modulo sD for every $s \geq 3$.

For triples containing $P_1 = \langle a, d, c \rangle \cong C_2^3$, use

$$(X,Y,Z) = (P_0, P_1, P_2),\ (P_0, P_1, P_3),\ (P_2, P_1, P_3).$$

Indeed $P_1P_2 = P_2 \cup dP_2$ and $P_1P_3 = P_3 \cup cP_3$. The extra representative belongs to $X \cap Y$ in each chosen triple, so IP immediately gives (3).

For $(X, Y, Z) = (P_0, P_2, P_3)$ the complete finite S_4 coset decomposition is

$$P_2P_3 = P_3 \cup cP_3 \cup bcP_3 \cup abcP_3.$$

The first three representatives lie in $P_0 \cap P_2$, so their intersections with P_0 are in the required product. The last coset has empty intersection: P_3 fixes $(0, 0)$ in the proved action, whereas abc sends it to $(1, 0)$. Its P_0-orbit is exactly $\{(0, 0), (3, 0), (2, 0), (1, -1)\}$, by the three explicit generator formulas. For $s \geq 3$ this excludes $(1, 0)$. Hence (3) holds for every triple of maximal parabolics.

For completeness these global and local triple properties imply full four-coset Helly. Normalize one coset to P_0. Its intersections with the other three cosets are cosets in P_0. Any two meet by the global triple property, and the local triple property in P_0 gives a common point of all four. Smaller families are already covered. UA.Geometry therefore gives thinness, residual connectivity and flag transitivity, and identifies E_s with the full type-preserving automorphism group, acting regularly on chambers.

Theorem 1. *For every $s \geq 3$, the literal presentation in [1, Theorem 3.1] defines E/sD of order $48s^3$. Its coset system is the universal locally toroidal regular rank-four hypertope specified there, with exact toroidal parameters $(2, 0)$ and $(s, 0)$ and the source's faithful degree-$4s$ CPR graph.*

Universality means the exact diagram presentation with the stated local relations: Lemma 1 proves a marked isomorphism, not merely the existence of a finite image.

5 The square family and its affine group

We use column coordinates and composition $(A, t)(B, u) = (AB, t + Au)$. The source writes Euclidean transformations on the right. Reversing an involutory word takes its inverse, so its order and its defining power relation are unaffected by this convention. Let $e_1, \ldots, e_4$ be the standard Euclidean basis, and put

$$h = \tfrac{1}{2}(e_1 + e_2 + e_3 + e_4), \qquad \Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}e_3 + \mathbb{Z}h.$$

All integer vectors below are coordinates in the ordered basis (e_1, e_2, e_3, h) of Λ.

Take the five source reflections, here specified in Euclidean coordinates:

$$\begin{aligned}
R_0(x) &= (1 - x_1, x_2, x_3, x_4),\\
R_1(x) &= x - \tfrac{1}{2}(x_1 - x_2 - x_3 - x_4)(1, -1, -1, -1),\\
R_2(x) &= (x_1, x_2, x_3, -x_4),\\
R_3(x) &= (x_1, x_2, x_4, x_3),\\
R_4(x) &= (x_1, x_3, x_2, x_4).
\end{aligned}$$

Set $E = \langle R_0, \ldots, R_4 \rangle$ and mark it by

$$t_0 = R_1, \quad t_1 = R_2, \quad t_2 = R_1R_3, \quad t_3 = R_0R_4. \tag{S.1}$$

These transformations preserve Λ. They generate E: besides $R_1 = t_0$ and $R_2 = t_1$, one has

$$R_3 = t_0t_2, \qquad R_4 = (t_0t_3)^3, \qquad R_0 = t_3(t_0t_3)^3. \tag{S.2}$$

The last two identities follow from the commuting pair R_0, R_4, the commuting pair R_1, R_4 and $(R_1R_0)^3 = 1$.

For $J \subseteq \{0,1,2,3\}$ write $E_J = \langle t_j : j \in J \rangle$. Let Q_J be its linear image and K_J its translation kernel. The full linear image Q has order 1152. A faithful finite action for its calculation is the action on the following 48 Euclidean roots:

$$\{\pm e_i\} \ \cup \ \{\tfrac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)\} \ \cup \ \{\pm e_i \pm e_j : i < j\}. \tag{S.3}$$

The roots span the space, so fixing them all forces a linear map to be the identity.

The exact affine computation gives the following complete table. The two nonsingleton translation lattices have bases displayed below it.

J	$\lvert Q_J\rvert$	K_J	J	$\lvert Q_J\rvert$	K_J
$\varnothing$	1	0	$\{3\}$	2	0
$\{0\}$	2	0	$\{0,3\}$	12	0
$\{1\}$	2	0	$\{1,3\}$	4	0
$\{0,1\}$	6	0	$\{0,1,3\}$	48	0
$\{2\}$	2	0	$\{2,3\}$	6	0
$\{0,2\}$	4	0	$\{0,2,3\}$	36	0
$\{1,2\}$	12	0	$\{1,2,3\}$	36	K
$\{0,1,2\}$	48	0	$\{0,1,2,3\}$	1152	$\mathbb{Z}^4$

Here

$$K = \mathbb{Z}u + \mathbb{Z}v, \qquad u = (1,-1,1,0), \quad v = (0,-1,0,1). \tag{S.4}$$

In particular the full translation lattice of E is exactly Λ. The entries are obtained by enumerating each finite Q_J, retaining affine word lifts and computing all normalized Schreier translations, as in AL.1 of the common paper. Integer Hermite reduction computes their entire generated lattice. It is essential here to use normalized transitions: a list of unnormalized differences alone would not suffice.

The table also exhibits an integral direct complement to K. Writing f_i for the integer coordinate vectors, the four columns u, v, f_2, f_3 form a unimodular matrix. More explicitly, every $x = (x_1, x_2, x_3, x_4) \in \mathbb{Z}^4$ has the unique expression

$$x = x_1 u + x_4 v + (x_2 + x_1 + x_4) f_2 + (x_3 - x_1) f_3. \tag{S.5}$$

Consequently $K \cap s\mathbb{Z}^4 = sK$ for every positive integer s.

6 The presentation and the period

Put $a = t_0, b = t_1, c = t_2, d = t_3$. Let $\mathcal{U}$ be the abstract group on four generators with their involution relations and

$$\begin{aligned} (ab)^3 = (ac)^2 = (ad)^6 = (bc)^6 = (bd)^2 = (cd)^3 = 1, \\ \big(c(da)^2\big)^2 = \big(b(ad)^2a\big)^2 = \big(a(bc)^2b\big)^2 = 1. \end{aligned} \tag{S.6}$$

These are the square diagram relations and the three fixed toroidal residue relations. All hold in the affine transformations. The marked map $\mathcal{U} \to Q$ is onto. Its exact Reidemeister–Schreier presentation, followed by Tietze simplification, has four kernel generators and the following six relators:

$$\begin{aligned} &z_4^{-1} z_1^{-1} z_4 z_1, \quad z_2 z_1^{-1} z_2^{-1} z_1, \quad z_4^{-1} z_2 z_4 z_2^{-1}, \\ &z_3 z_1 z_3^{-1} z_1^{-1}, \quad z_3^{-1} z_4^{-1} z_3 z_4, \quad z_3^{-1} z_2 z_3 z_2^{-1}. \end{aligned} \tag{S.7}$$

Thus the actual kernel, not merely its abelianization, is free abelian of rank four. The affine translation kernel has the same rank by the preceding table. AL.2 now identifies $\mathcal{U}$ with E, with the marking preserved.

Define $w = d(cb)^2$. Direct affine multiplication gives

$$w^2 = (I, v). \tag{S.8}$$

The Q-orbit of v consists of 24 vectors and spans $\mathbb{Z}^4$ over the integers. This is checked by its full integral Hermite basis I_4. Conjugating a translation by an affine transformation applies its linear part to the translation vector. Hence the normal closure of w^{2s} in E is exactly $s\Lambda$, and

$$\mathcal{U}/\langle\langle w^{2s}\rangle\rangle \cong E/s\Lambda, \qquad |E/s\Lambda| = 1152s^4. \tag{S.9}$$

This proves a marked presentation identity for all s at once.

For comparison with the source's original toroid, the five-generator Coxeter presentation $[3, 3, 4, 3]$ has the same affine realization. The separate exact kernel calculation over Q again reduces to four generators with all six pairwise commutator relations and no others. AL.2 identifies that presentation with E. Its period word satisfies

$$\big(R_0R_1(R_2R_3R_4)^3\big)^2 = (I, f_4).$$

The Q-orbit of f_4 also spans $\mathbb{Z}^4$. Its sth power therefore imposes exactly $s\Lambda$. Together with (S.2), this proves that the groups in Proposition 4.1 are the same marked constructions under the displayed change of generators, not just groups of the same order.

7 The ordered residues

Let P_i be the image of $E_{I\setminus\{i\}}$ in $E/s\Lambda$. For $i = 1, 2, 3$ the linear maps are injective on the entire affine parabolics, so their groups do not collapse in any translation quotient. In the respective ordered paths

$$(t_2, t_3, t_0), \qquad (t_1, t_0, t_3), \qquad (t_0, t_1, t_2),$$

the edge orders are $3, 6$ and the end generators commute. Their additional relations in (S.6) are precisely those of $\{3, 6\}_{(1,1)}, \{3, 6\}_{(2,0)}, \{3, 6\}_{(2,0)}$. The standard triangle presentations with these relations have orders $36, 48, 48$, respectively; exact finite presentation calculations give these orders. The table gives the same orders for the three images, so the marked epimorphisms are isomorphisms. No classification by order alone is being used.

For P_0, put $(A, B, C) = (t_3, t_2, t_1)$ and let

$$\Delta = \langle A, B, C \mid A^2 = B^2 = C^2 = (AB)^3 = (BC)^6 = (AC)^2 = 1\rangle.$$

Its epimorphism onto the actual linear group $Q_{\{1,2,3\}}$ of order 36 has kernel with the exact simplified presentation

$$\langle z_1, z_2 \mid z_2z_1^{-1}z_2^{-1}z_1 = 1\rangle.$$

The affine image has translation lattice K of rank two. The restriction from this kernel onto K is a surjection of free abelian groups of the same rank, hence an isomorphism. This is the same rank argument as AL.2, now with rank two inside a four-dimensional ambient affine space. Thus $E_{\{1,2,3\}}$ is the marked triangle group Δ itself.

The $Q_{\{1,2,3\}}$-orbit of v has six vectors and spans exactly K. It follows that imposing $(A(BC)^2)^{2s} = 1$ on Δ factors its translation lattice by sK. By (S.5) the ambient quotient induces exactly the same subgroup sK, without any extra residue collapse. This is the standard marked presentation of $\{3, 6\}_{(s,s)}$, and its order is $36s^2$. The ordered rank-three groups are therefore exactly the four claimed in Proposition 4.1.

The source also computes two word orders to distinguish the possible residue periods. Here (S.8) has order s, while $A(BC)^2B$ has linear part of order three and cube (I, u). The primitive vector u has order s modulo $s\Lambda$, so this word has order $3s$. The word itself is generally not a translation: it is its cube that is a translation. This corrects the terminology in the source without changing either asserted order.

8 Intersections and flag transitivity for every period

The fixed linear marking has the intersection property. The certificate checks all 16^2 equalities $Q_J \cap Q_T = Q_{J\cap T}$ as literal sets of matrices. Its four maximal parabolic coset sets have $32, 32, 24, 24$ members. Complete enumeration of pairwise incident cosets gives the following numbers after translating the first coset to the subgroup itself:

types	012	013	023	123	0123
pairwise incident families	18	18	18	18	36.

Every listed family has a common element. Rank-one and rank-two families already satisfy that assertion by nonemptiness and incidence. This proves typed coset Helly for the fixed linear quotient. The normalization loses no family, because right translation acts transitively on the cosets of the first type.

Now fix a positive integer s and set $G = E/s\Lambda$, with the induced map $q : G \to Q$. Its kernel is $N = \Lambda/s\Lambda$. For a proper marked parabolic G_J, the table and the fact that $s\Lambda$ consists of translations give

$$N \cap G_J = \begin{cases} (K + s\Lambda)/s\Lambda, & J = \{1, 2, 3\}, \\ 1, & \text{every other proper } J. \end{cases} \tag{S.10}$$

For the full index set the intersection is N. This entire kernel profile has the intersection property: two different proper supports cannot both have a nontrivial kernel intersection. UA.K therefore proves the intersection property in G.

Equation (S.5) yields the internal direct product

$$N \cong (K/sK) \ \oplus \ \big((\mathbb{Z}f_2 + \mathbb{Z}f_3)/s(\mathbb{Z}f_2 + \mathbb{Z}f_3)\big).$$

Each maximal parabolic intersection in (S.10) is a product of some of these two factors. The linear quotient has IP and typed coset Helly, so UA.SK.Helly transfers typed coset Helly to G. Its four markers have order exactly two, as their nontrivial linear images do. UA.Geometry now proves that the coset incidence system is thin, residually connected and flag-transitive, and that its full type-preserving automorphism group is G.

Finally (S.6) and (S.9) give exactly the diagram relations and the four ordered residue relations. There is no extra global relation. This proves universality, completing Theorem 4.3 for its full stated range $s \geq 2$. The argument in fact also verifies these group and coset assertions at $s = 1$; no additional claim about the terminology for the source's rank-five toroid at that parameter is needed.

9 The literal presentations and a small countermodel

Here the order of the markers is $a = \rho_0, b = \rho_1, c = \rho_2, d = \rho_3$. Let B denote the following thirteen words, all imposed as relators:

$$a^2, b^2, c^2, d^2, (ab)^3, (ac)^6, (ad)^2, (bc)^2, (bd)^6, (cd)^3,$$
$$(d(ca)^2c)^2, \quad (a(bd)^2)^2, \quad (b(ac)^2a)^2.$$

Put $w = c(db)^2d$ and $v = c(db)^2$. For $t \in \{1, 2\}$ define

$$P_t = \langle a, b, c, d \mid B, w^{3t} \rangle, \qquad V_t = \langle a, b, c, d \mid B, v^{2t} \rangle. \tag{F.1}$$

Thus P_1, P_2 are the original presentations. The printed replacement on p. 10 has an extra closing parenthesis, but removes the final d and uses exponent two; p. 11 unambiguously uses v^4. The assertions that these replacements imply $(bd)^2 = 1$ and produce the stated 240-chamber polytope are false for these literal words.

Here is one countermodel for both, in the symmetric group on eleven points; multiplication is functional composition, right factor first:

$$a = (1\,3)(2\,4)(5\,6)(7\,9), \qquad b = (1\,2)(3\,5)(4\,6)(7\,8),$$
$$c = (3\,6)(4\,5)(9\,10), \qquad d = (1\,4)(2\,3)(10\,11).$$

Direct substitution makes every word of B, v^2, and v^4 the identity, while bd has order six. In particular $(bd)^2$ sends 1 to 4. Closing these four permutations under multiplication gives 720 distinct elements. Hence neither V_t can have order 240: each has a quotient of order 720. The complete integer-permutation check is `finite/countermodel.py`. Six declarations in `finite/formal/SectionFive.lean` prove the thirteen shared relations, both replacement relations, the failure of $(bd)^2 = 1$, and both resulting universal nonconsequences. The order 720 computation is not included in the Lean scope.

10 Exact corrected groups and ordered residues

Within P_t, for an omitted index i let G_i be the subgroup generated by the other three markers. We write $T(s, 0)$ for the marked triangular torus group with presentation

$$\langle x, y, z \mid x^2, y^2, z^2, (xy)^3, (yz)^6, (xz)^2, (x(yz)^2y)^s \rangle,$$

and $T(t, t)$ for the same first six relators followed by $(x(yz)^2)^{2t}$. These are the usual triangle-reflection torus conventions in [1]; their orders are $12s^2$ and $36t^2$.

Theorem 2. *For $t = 1, 2$, the presentations P_t and V_t are isomorphic by the map fixing all four named generators, and $|P_t| = 720t^6$. Their ordered maximal parabolics are as follows.*

Omitted index	*Ordered markers (x, y, z)*	*Marked group*	*Order*
0	(c, d, b)	$T(t, t)$	$36t^2$
1	(d, c, a)	$T(2, 0)$	*48*
2	(a, b, d)	$T(1, 1)$	*36*
3	(b, a, c)	$T(2, 0)$	*48*

The marked map $T(3t, 0) \longrightarrow G_0$ is an epimorphism with kernel of order three. The associated torus-map covering has degree three.

Proof. The complete required program `finite/exact_groups.g` constructs each abstract group in (F.1) and calls GAP's exact fp-group isomorphism routine, not just a homomorphism onto an assumed finite image. It obtains orders 720 and 46080. In the original presented group it verifies $v^{2t} = 1$; in the modified presented group it verifies $w^{3t} = 1$. Thus the two identity-on-generators homomorphisms exist and are inverse. Exact GAP fp-group and coset computations are declared computational premises of this proof.

For each row of the table, the program constructs the displayed three-generator abstract torus presentation, checks its order, and constructs a homomorphism to the actual parabolic with exactly the listed generator images. This map is onto and both groups have the listed order, so it is a marked isomorphism, not identification from orders alone. The analogous map from $T(3t, 0)$ is onto G_0; its domain order $108t^2$ gives kernel size three. These are the standard triangular translation lattices, with no local rank-two collapse: each relevant marked dihedral subgroup has its prescribed order. Hence the torus map is a covering of degree three.

For a second, elementary finite check, `finite/model_data.json` records faithful images on 13 and 30 points. `finite/finite_models.py` closes these literal permutations, substitutes every original and modified relator, and verifies all parabolics and all tests below. Its closure gives the finite-image lower bound only; the abstract upper bound still comes from the separate exact fp-group computation above. □

11 Intersection, flags and universality

Lemma 3. *For three subgroups H, K, L of a group, all pairwise intersecting triples of their right cosets have a common point if and only if*

$$H \cap KL = (H \cap K)(H \cap L). \tag{F.2}$$

If four subgroup types satisfy this for every triple, and the three intersection subgroups inside any one type also satisfy it, every pairwise intersecting four-coset family has a common point.

Proof. Translate an intersecting pair of cosets to H, K. Choose $h \in H \cap Lx$ and $k \in K \cap Lx$. Then $hk^{-1} \in L$, so $h \in H \cap LK$. Taking inverses in (F.2) gives $h = l_0 k_0$ with $l_0 \in H \cap L$ and $k_0 \in H \cap K$. Now $k_0 = l_0^{-1} h \in Lx$ is a common point. Conversely the common-point property applied to these normalized triples gives the factorization. For four cosets, intersect each of the other three with a fixed fourth. Global triple intersection makes these three cosets inside the fourth pairwise intersecting. The local three-coset property supplies their common point. □

For each of the two actual finite groups, both required checks enumerate all 16 generated subgroups G_J, including $G_\varnothing = 1$, and test all 256 equations

$$G_J \cap G_K = G_{J \cap K}. \tag{F.3}$$

They also test (F.2) for all four triples of maximal parabolics, and for the three subgroups $G_i \cap G_j$ inside each G_i (four more tests). All equalities hold as full sets, not merely as cardinalities. Lemma 3 therefore proves the coset geometry flag-transitive in every rank up to four.

The markers are nontrivial involutions. Equation (F.3) and the full typed coset Helly property therefore satisfy UA.Geometry of the common paper. The coset system is thin, residually connected and flag-transitive, and the displayed group is its full type-preserving automorphism group.

The modified presentation V_t comprises precisely the circuit relations and the four displayed ordered rank-three period relations. By the marked isomorphism, P_t is therefore also the universal marked group with those local presentations. In particular an alternative compatible marked regular geometry is a quotient; the local maps are isomorphisms when its parabolics have those exact presentations. Neither literal replacement changes this group. The distinction between ordered local constraints and an unordered list of toroidal maps is retained throughout; we do not claim to classify every permutation of the local data.

12 The relation with the full cubic toroid

Let Q be the signed permutation group on three coordinates and $D = \{v \in \mathbb{Z}^3 : v_1 + v_2 + v_3 \equiv 0 \pmod 2\}$. In the notation of the reviewed cubic-family proof, the hypertope group is $E_s = E/sD$, where

$$E = \{(q, v) \in Q \ltimes \mathbb{Z}^3 : \textstyle\sum v_i \equiv \epsilon(q) \pmod 2\}, \qquad |E_s| = 48s^3.$$

The introductory sentence on page5 of [1] identifies this group with the full group of the cubic toroid $\{4, 3, 4\}_{(s,s,0)}$. The earlier account [2, pp. 14–15] correctly describes an index-two subgroup. Here is the exact change of generators and coordinates.

The standard Euclidean realization of $[4, 3, 4]$ is generated by

$$R_0(x) = (1 - x_1, x_2, x_3), \qquad R_1(x) = (x_2, x_1, x_3),$$
$$R_2(x) = (x_1, x_3, x_2), \qquad R_3(x) = (x_1, x_2, -x_3).$$

Its affine image is $W = Q \ltimes \mathbb{Z}^3$: the last three generators give Q, and multiplication of R_0 by the corresponding linear reflection gives translation by e_1. Conjugating gives all coordinate

translations. The faithful Euclidean Coxeter realization is a retained classical fact. Direct multiplication gives

$$(R_0R_1R_2R_3R_2)^2 = (I, e_1 + e_2).$$

Its Q-orbit generates D. Thus the additional relation with exponent $2s$ used in [2] imposes precisely sD, and the full toroid group is W/sD, of order $48[\mathbb{Z}^3 : sD] = 96s^3$.

Put $p = (1/2, 1/2, 1/2)$ and $T(x) = \mathrm{diag}(1, -1, -1)(x - p)$. Conjugating the four displayed ambient words by T gives

$$\begin{aligned} T(R_0R_1R_0)T^{-1}(x) &= (x_2, x_1, x_3), \\ TR_2T^{-1}(x) &= (x_1, x_3, x_2), \\ T(R_1R_3)T^{-1}(x) &= (-x_2, -x_1, 1 - x_3), \\ TR_1T^{-1}(x) &= (-x_2, -x_1, x_3). \end{aligned}$$

These are exactly the four generators of E in source-label order. The linear part of T preserves D, so this conjugacy descends to the period quotients. Consequently the displayed subgroup of W/sD is isomorphic to E/sD and has index two. In [2] the Petrie generators are $\alpha_0 = R_0$, $\alpha_1 = R_1$, $\alpha_2 = R_2$ and $\alpha_3 = R_1R_3$; its four subgroup generators are the same set, ordered here as $(\alpha_1^{\alpha_0}, \alpha_2, \alpha_3, \alpha_1)$ to match [1].

The family still arises from the stated toroid. The correction is that its group is the displayed index-two subgroup, rather than the full toroid group. This changes no principal result of Section3 of [1]. It does not claim a new family or a resolution of an open question.

13 A smaller faithful permutation action

After the drawings, [1, p. 6] suggests that $4s$ is "most probably" minimal. This is not a proved assertion. In that generality it fails: $3D \cap 5D = 15D$ gives an injection $E/15D \longrightarrow E/3D \times E/5D$. The disjoint union of the two faithful actions has degree $12 + 20 = 32 < 60$. This is an upper bound, not a classification of minimum degrees. We make no claim of historical priority or of resolving a question that is currently open.

14 Attribution and verification

The source's two families, their intended residue types and their permutation graphs are prior results. The affine generators and parameter lattices are also prior material. Our proof identifies the abstract groups with these affine models through AL.2, computes the complete integral parabolic lattices through AL.1, and applies the common kernel and coset theorems. For the square family, a single fixed linear quotient and a direct summand of the translation lattice replace the parameter-dependent subgroup and coset arguments. The cubic flag-transitivity argument retains the source's four-coset decomposition; the affine model and its faithful action make its intersection and orbit assertions explicit. The common theorems, these applications and the corrections are distinguished in the source map. The degree 32 action at parameter 15 refutes the source's unrestricted minimum-degree suggestion; it is intransitive and says nothing about the minimum transitive degree. The distinction between the full cubic toroid and its index-two subgroup is already explicit in [2]. The two finite replacement claims are corrected for the literal words in the retained source edition. None of these corrections is assigned historical priority or current open-conjecture credit.

The complete required programs and literal data are retained in the accompanying directories. GAP's exact presentation algorithms and the integer matrix and permutation calculations remain declared computational premises. The six Lean declarations in the finite case verify the shared relators, the two modified relators and the failure of the asserted consequence in

an explicit permutation model. They do not formalize the group orders, the infinite families or the geometry. The remaining formal verification is separate from the written proof and computational evidence.

References

[1] M. E. Fernandes, D. Leemans, C. A. Piedade and A. Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, retained twelve-page author edition.

[2] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, retained 21-page author edition, pp. 14–15.

[3] *Supported kernels, translation lattices and finite marked quotients*, accepted common manuscript, AL.1–AL.2, AA.ParabolicPeriods, UA.K, UA.SK.Helly and UA.Geometry.